# Designing and Evaluating Public-Facing Multimodal LLMs for Early Autism Screening.

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**Abstract.** Autism Spectrum Disorder (ASD) is a growing public health concern, yet early identification remains difficult, with many children diagnosed long after symptoms first emerge. Although videos capture rich early behavioral signals, their unstructured nature makes large-scale manual review impractical, and traditional machine learning approaches based on engineered features struggle with real-world variability. We design and evaluate a public-facing, multimodal large language model (Multimodal LLM) framework to support behavioral review in early autism screening using user-generated video data. Our prompt-driven approach integrates sampled video frames with transcribed audio to identify autism-related behaviors across naturalistic home recordings submitted by caregivers. The framework incorporates local deployment and human-in-the-loop controls that allow users to curate visual evidence and verify inputs, supporting interpretability and privacy preservation. Unlike end-to-end classifiers that operate as opaque decision pipelines, the framework produces structured, human-readable behavioral annotations grounded in observable evidence. The evaluation across ten autism-related behavioral categories using therapist-annotated videos shows strong performance for visually salient, structured behaviors (e.g., Object Lining-Up, $F1 = 0.81$) while exhibiting conservative, high-specificity behavior for linguistically and contextually complex categories (e.g., Non-Typical Language, $F1 = 0.15$) - a category-level variation that itself constitutes a key finding for future multimodal ASD screening research. Performance is interpreted in the context of substantial class imbalance, with F1 and specificity foregrounded over accuracy. These findings demonstrate the potential of Multimodal LLMs to function as a pre-referral behavioral triage layer, surfacing high-confidence signals from caregiver-submitted video without displacing clinical judgment, and establish a governance architecture for deploying citizen-facing AI in participatory public health workflows. The contribution is a replicable design model that responsibly integrates citizen generated data into early identification pipelines.

## 1. Introduction

Autism Spectrum Disorder (ASD) is increasingly recognized as a major public health and educational concern due to both rising case numbers and persistent challenges in early identification. The 2022 Autism and Developmental Disabilities Monitoring Network's national surveillance showed that the overall prevalence rate for 8-year-old children was 32.2 per 1,000 children (about 1 in 31), as reported by (Shaw et al., 2025). Despite this, the average age of diagnosis has remained high. Shaw et al. (2022) found that while the age of diagnosis for children with clear records decreased slightly to 54 months by 2016, it remained at 62 months when including children who met surveillance criteria but were missed by the system.

Public health professionals identify the promotion of early diagnosis as a critical need, yet many children are not diagnosed until well after symptoms become distinguishable (Wallace et al., 2012). Early detection enables personalized interventions to address social communication and behavioral issues, allowing families

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access to support services and educational resources that lead to better coping mechanisms and decreased parental stress (Okoye et al., 2023; Vu et al., 2023; Mandy et al., 2022; French et al., 2023). In addition to that, delays in identification result in significantly higher healthcare resource utilization and economic burdens during the diagnostic journey (Matin et al., 2022). In this context, empowering caregivers to participate actively in early behavioral documentation is essential to bridging the gap between symptom emergence and formal diagnosis - not by replacing clinical tools, but by generating timely, structured evidence that supports referral decisions. To operationalize this participation, the public must be equipped with the specific ability to recognize 'red flags' and developmental milestones, ensuring that initial concerns translate effectively into timely professional referrals (Wallace et al., 2012; Issac et al., 2025).

While caregiver and citizen-generated data play a vital role in capturing early behavioral red flags, their high volume and unstructured nature make manual review for identifying ASD-related behavioral traits time-consuming, subjective, and impractical at scale (Washington et al., 2021). To address these challenges, Kim et al. (2025) introduced a machine learning-based approach for the early identification of ASD, demonstrating the potential of AI to support behavioral screening. However, traditional machine learning approaches rely on predefined engineered features and are often not well-suited to the variability and noise of user-generated video data. Realization of the full potential of these technologies requires moving beyond single-modality or structured assessments to leverage the richness of 'in-the-wild' data. Expanding on this foundation, our study utilizes multimodal and large language models to process the complex, unstructured audio-visual cues found in user-generated videos, which are an emerging source of observational data, through everyday digital platforms that often capture behaviors prompting early concern.

This paper aims to examine the effectiveness and governance implications of deploying multimodal LLMs as an early autism screening support tool. We position AI not as a diagnostic instrument but as a structured evidence curator that lowers the barrier to timely professional engagement. Specifically, we aim to 1) evaluate how accurately a Multimodal LLM classifies autism-related behavioral categories from user-generated data, and 2) explore design implications for deploying such a system responsibly in community and public-sector contexts. To answer these questions, we introduce a prompt driven, Multimodal LLM framework for analyzing user generated video clips shared on social media platforms. A carefully engineered prompt directs the model to detect behaviors indicative of ASD in accordance with the clinical criteria outlined by (Deng et al., 2024). The prompt integrates multimodal inputs - audio and video. We first extract the audio track and transcribe it into text, thereby preserving spoken dialogue and salient environmental sounds. Subsequently, the system automatically samples candidate frames from the video stream. Users then employ a custom interface to manually select and curate the most pertinent frames from this set to be passed to the model. This ensures that the visual evidence sent to the system remains under human oversight. The textual transcription, the selected video frames, and the original prompt are concatenated into a single input that is forwarded to the Multimodal LLM system. Processing is performed by a locally deployed vision language model, *Gemma 3 27B*, which fuses visual information with the audio transcript to identify potential ASD related behaviors.

Our experiment results indicated that, across ten ASD-related behavioral categories, the Multimodal LLM achieved a macro-average F1 score of $0.49$ (precision: $0.60$, recall: $0.41$), reflecting considerable variation across behavior types. While the overall accuracy was $0.86$, this figure is inflated by pronounced class imbalance; negative examples dominate each behavioral class, making F1 and per-class specificity the more informative indicators of system performance. We find that while the model reliably flags visually salient behaviors, such as object lining or repetitive motor movements, it exercises caution with socially or emotionally sensitive categories, maintaining high specificity to minimize false alarms. These results suggest that Multimodal LLMs can reliably identify a subset of autism-related behavioral categories from user-generated video clips, particularly those with visually salient, spatially structured cues such as Object Lining-Up ($F1 = 0.81$), Aggressive Behavior ($F1 = 0.72$), and Upper Limb Stereotypies ($F1 = 0.66$). Performance is substantially lower for behaviorally nuanced or context-dependent categories such as Non-Typical Language (F1=0.15), Self-Hitting ($F1 = 0.26$), and Hyper- or Hyporeactivity to Sensory Input ($F1 = 0.30$), where the system defaults to conservative, high-specificity behavior. This category-level variation is itself a key finding, with implications for how such systems should be scoped and evaluated in practice. We further explain the model performance through an interactive user interface that permits users to manually select and curate the most pertinent video frames and verify the prompt text submitted to the Multimodal LLM. We then present the lessons learned from the selected instances and discuss how this proposed framework can maintain human oversight over any visual evidence transmitted to the system, as well as the broader implications of deploying such AI systems in real world ASD screening, including considerations of privacy preservation, interpretability, and user oversight.

This work contributes to the literature on Digital Government and AI by illustrating how Multimodal LLMs can be designed as public-facing, decision-support infrastructures that responsibly support early autism identification — specifically, by enabling caregivers to generate structured behavioral evidence that can accelerate clinical referral without substituting for it. Situated at the intersection of AI, public health, and education, the study extends digital government scholarship on participatory governance, civic data use, and human-centered AI by operationalizing caregiver- and citizen-generated video data as a scalable yet carefully governed resource for early behavioral identification. It advances AI research by demonstrating how prompt-driven, human-in-power Multimodal LLM can process unstructured, in-the-wild audio-visual data while maintaining conservative performance profiles aligned with public-sector values such as safety, accountability, transparency, and trust. By emphasizing local deployment, transparent reasoning, and human agency, this work shows how AI-enabled decision-support systems may reduce delays in early screening, empower caregivers as active participants in early behavioral flagging, and support more equitable and timely public health interventions within digitally mediated government and community ecosystems.

## 2. Literature Review

### *2.1. Multimodal and Caregiver-Informed ASD Detection*

Researchers have explored several approaches to identify autism spectrum disorder (ASD) early, including screening tools, clinical observations, and longitudinal monitoring of development. Although early detection is important, many children still wait a long time for diagnosis and follow-up care because there are not enough specialists, resources are unevenly distributed, and there are challenges in public health and education systems (Wallace et al., 2012; Daniels and Mandell, 2014; Shaw et al., 2022). These delays are significant because early diagnosis helps children receive focused support that can improve social communication, reduce caregiver stress, and lead to better long-term outcomes (Mandy et al., 2022; Okoye et al., 2023; Vu et al., 2023).

Recent research highlights the important role caregivers play in detecting early signs of autism. Parents and family members often notice unusual behaviors before a formal diagnosis, so their observations are valuable but are often ignored (Zwaigenbaum et al., 2015). Most public screening still relies on clinician-led tools and in-person assessments, so caregiver input is rarely used. This gap has generated interest in new technologies that connect caregiver knowledge with formal assessments, especially in areas where specialists are difficult to reach. At the same time, digital public health research has examined how citizen-generated unstructured data, such as social media posts, images, and videos, can support health monitoring and behavior tracking. Shakeri Hossein Abad et al. (2021) found that these sources can provide public health insights more quickly and easily, especially for behavioral and mental health issues. Other studies show that social media text can expose patterns related to autism traits and other mental health conditions (De Choudhury et al., 2013). However, the use of citizen-generated data creates challenges for interpretation, transparency, and responsible use in public decision-making.

Early research in this field mainly focused on text and missed important nonverbal behaviors. Yet, autism screening depends heavily on visual and hearing cues such as gaze, body movements, repetitive actions, and social engagement. To address this, recent studies have started using video-based and multimodal methods (Kim et al., 2025; Dos Santos Melício et al., 2025). Video lets researchers analyze both visual and audio signals together, giving greater context than text alone. Kommineni et al. (2025) showed that multimodal models can detect complex patterns in autism-related videos. Earlier computer vision studies also looked at automated evaluation of facial expressions, gaze, and movement for ASD screening (Hashemi et al., 2021; Washington et al., 2020). Deng et al. (2024) introduced the AV-ASD dataset - the same annotated corpus used in this study - and demonstrated that combining audio and visual inputs improves recognition of autism-related behavioral signals compared to single-modality approaches. Their framework, however, relies on supervised deep learning models trained specifically on their annotated data, positions the system as a researcher- or clinician-facing pipeline, and does not address usability, user agency, or deployment in public-sector contexts. Kommineni et al. (2025) extended the use of multimodal foundation models to autism diagnostic video analysis, demonstrating that large pretrained models can generalize to ASD-related behavioral understanding. Their evaluation, however, focuses on structured clinical recordings from the ADOS-2 assessment protocol, a controlled, expert-administered diagnostic instrument, rather than unstructured, citizen-generated social media content. The present study is neither a direct continuation of either work nor a replication. It extends the shared multimodal direction by: (1) applying a prompt-driven, zero-shot LLM framework that requires no

task-specific training; (2) evaluating on user-generated, in-the-wild video data using the Deng et al. (2024) dataset as a benchmark; and (3) introducing a public-facing, locally deployed system with human-in-power controls designed for caregiver and community use - dimensions absent from both prior works.

Even with these technical advances, most video-based autism analysis systems are designed for researchers or clinical experts and emphasize performance over usability, clarity, or practical use. Sohn et al. (2025) found that while AI autism tools can be accurate in regulated environments, few are intended for large-scale use, good governance, or public-sector needs. In practice, these systems often function as black boxes in the operational sense: they produce numeric predictions or class labels with no accompanying rationale, offer no mechanism for users to inspect or curate the inputs driving those predictions, and provide no natural-language explanation accessible to non-specialists. This limits trust and interpretability for caregivers, teachers, and public health workers who must act on the outputs. Recent advances in generative AI, especially vision-language models (VLMs) - a key class of Multimodal LLMs - offer ways to solve these problems. Unlike older models, VLMs can create natural-language descriptions, summarize complex visual scenes, and answer user questions (Yoo et al., 2026). This helps people interpret information rather than just receiving unclear predictions (Hartsock and Rasool, 2024). In healthcare and other sensitive fields, most agree that AI should support, rather than replace, human assessment, with an emphasis on transparency, accountability, and user control (Holzinger et al., 2019).

### *2.2. Human-Centered AI in Public Health Screening*

Public sector AI deployment introduces demands that go beyond technical performance. In e-government and digital public service contexts, technology must serve citizens who did not choose the system, may not understand its logic, and cannot easily opt out of its consequences - conditions that make transparency, accountability, and user agency structural requirements rather than design preferences. Healthcare represents a particularly high-stakes instance of this: when AI is introduced into early autism screening pathways, it enters a space already shaped by institutional power asymmetries, caregiver vulnerability, and long histories of delayed or inequitable access to diagnosis (Daniels and Mandell, 2014; Wallace et al., 2012). Tools designed for researchers or clinicians, however technically capable, tend to reproduce rather than redistribute these asymmetries when they are deployed without mechanisms for citizen participation or interpretive control. This study responds to that structural gap by designing for caregivers as primary users - not as passive inputs to a diagnostic pipeline, but as active curators of the evidence that shapes it.

Research on human-centered AI and human-AI interaction supports this view. Design guidelines state that good AI systems should let users remain in control, provide clear explanations, and involve people in decision-making (Amershi et al., 2019). Building on this construct, new ethical and administrative frameworks suggest moving from human-in-the-loop models, where humans ratify machine outputs after the fact, to human-in-power models, where users exercise genuine authority over the evidentiary process-choosing what the system sees and retaining interpretive responsibility for what it concludes (Zheng et al., 2024). The distinction is more than terminological. In sensitive public-sector contexts, human-in-the-loop designs can create a false sense of oversight: a caregiver who reviews a model's output without controlling its inputs has limited ability to contest or contextualize its conclusions. Human-in-power design restores that ability by making user curation structurally prior to model inference. This shift also aligns with the concept of IT mindfulness - the extent to which users engage with technology attentively, openly, and in a non-automatic manner to new information rather than operating on automatic pilot (Thatcher et al., 2018). The frame-selection protocol operationalizes IT mindfulness by design: rather than passively receiving a prediction, users actively attend to the visual evidence, exercising judgment about which moments are clinically meaningful before the model ever processes them.

At present, a disconnect remains between high level principles of user agency and their realization in functional public sector tools; although the risks of opaque automation are well documented, empirical frameworks that successfully operationalize user agency within the technical constraints of multimodal autism screening are scarce. To address this gap, we adopt a Design Science Research (DSR) approach—viewing the construction and evaluation of purposeful artifacts as a legitimate form of knowledge production in information systems (Hevner et al., 2004)—and test the practical viability of a human guided, prompt driven analysis system. Rather than merely testing a hypothesis about an existing phenomenon, DSR asks whether a designed artifact can solve a real world problem and what generalizable design knowledge it yields. In our study, the artifact is a prompt driven multimodal LLM framework for public facing behavioral review, whose

evaluation against therapist annotated video data constitutes the DSR evaluation cycle. By providing concrete evidence that manual evidence curation using citizen generated data can stabilize model performance, we aim to establish generative AI as a transparent decision support tool. The design principles derived from this evaluation—pertaining to human in the loop controls, prompt driven behavioral operationalization, and conservative operating profiles for sensitive public health contexts—constitute the theoretical contribution (Gregor and Hevner, 2013). Following Gregor and Hevner's (2013) knowledge contribution framework, this work represents an exaptation contribution: it applies a known class of artifact (multimodal LLMs) to a new problem context (public facing, caregiver operated behavioral review in early autism screening), with the primary theoretical value residing in the design principles emerging from this novel application rather than in the artifact technology itself.

## 3. Methodology

### *3.1. Data Collection*

We collected the therapist-annotated dataset from (Deng et al., 2024). The original dataset contains 928 video clips from 569 unique videos across 10 categories collected from social media platforms (YouTube and Facebook). The dataset was annotated by the research team, validating the presence/absence (1/0) of 9 distinct autistic behavioral categories: *Absence or Avoidance of Eye Contact*, *Aggressive Behavior*, *Hyper- or Hyporeactivity to Sensory Input*, *Non-Responsiveness to Verbal Interaction*, *Non-Typical Language*, *Object Lining-Up*, *Self-Hitting or Self-Injurious Behavior*, *Self-Spinning or Spinning Objects*, *Upper Limb Stereotypies*. When none of the other behaviors were significant, the clips were assigned the default category of *Background*. We compiled our dataset by retrieving video samples from the sources using the video links provided in the GitHub repository (Collected in July, 2025). Out of the original 929 clips, 788 could be retrieved because the remaining files were no longer available. We followed the video extraction script from the repository to extract the segments from the original videos; all extracted clips were between 5 and 100 seconds in length.

### *3.2. Data Preprocessing*

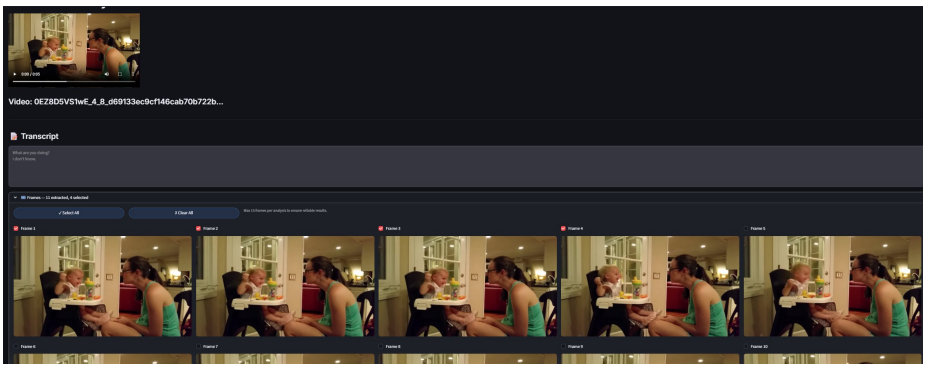



**Fig. 1** – Human-in-the-loop frame selection interface.

To facilitate multimodal analysis, we decomposed the raw video clips into distinct textual and visual streams.Audio components of the clips were extracted using *OpenAI Whisper* and converted to text files (Radford et al., 2022). OpenAI Whisper was used due to its proven robustness in handling the significant background noise and variable recording quality inherent to in-the-wild social media data. For the video component, we used *OpenCV cv2 (version 4.13.0)* to convert the video to sequential images (Bradski, 2000). OpenCV (cv2) is used to ensure deterministic, artifact-free frame extraction that preserves the raw pixel fidelity of transient behavioral signals without the smoothing effects of deep-learning decoders. We employed a human-in-the-loop frame selection protocol with the assumption that automated sampling may miss transient behavioral cues. For each video clip, a human reviewer used a custom interface to scrub through the footage and manually select up to $12$ frames that best captured the onset, peak, and offset of the target behavior. The number of frames selected ($N_i$) was variable, allowing the user to curate only the necessary visual evidence for each specific instance, whether that required $5$ frames for a brief gesture or $12$ for a complex interaction. While manual frame selection enables deliberate evidence curation, the system also supports full-frame submission via a "Select All" control, allowing users to trade curation granularity for speed when reviewing lower-priority

or time-constrained clips. This flexibility partially mitigates scalability concerns while preserving user agency over the choice itself. This approach mimics the proposed real-world workflow where a caregiver would curate evidence rather than submitting raw video dumps. These human-selected frames constituted the visual input for the Multimodal LLM ($\{v\}_{i=1}^{N}$).

It should be noted that this curation model assumes a baseline familiarity with ASD behavioral cues that non-specialist caregivers may not possess. To operationalize the human-in-power design for public deployment, the interface would require an embedded knowledge scaffolding layer — for example, a pre-review prompt showing annotated examples of each behavioral category — so that caregivers can make informed frame selections without prior clinical training. This scaffolding does not alter the AI system's architecture but is a necessary design component for any public-sector deployment and is proposed here as a specification for future implementation.

### *3.3. System Implementation*

We deployed the **Gemma 3 27B Instruction-Tuned** model locally using the Ollama inference framework (Team, 2025). We utilized 4-bit quantization **(q4_k_m)** to enable efficient inference on consumer-grade hardware. While 4-bit quantization is an aggressive precision reduction, prior work has shown that modern k-quant schemes applied to large-parameter models ($\geq 27B$) retain the majority of full-precision task performance, with observed degradation typically under 2–5% on reasoning and classification benchmarks compared to BF16 baselines (Dettmers et al., 2023). The q4_k_m scheme specifically applies higher precision to the attention and feed-forward layers most critical to output quality, partially mitigating precision loss inherent to 4-bit representation. The model was accessed via Ollama's local REST API to ensure data privacy and consistent latency. By restricting data processing to the local network, we overcame a primary barrier to public sector adoption of cloud-based AI because sensitive video data never leaves the secure local environment.

### *3.4. Prompt Design*

To ensure reproducibility, we configured the model with $temperature = 0$ and a fixed random seed ($seed = 42$), which produces greedy decoding behavior in the Ollama inference framework. Identical outputs were confirmed empirically across a five-clip validation set, verifying deterministic behavior under this configuration. While $top_k = 1$ was initially evaluated to enforce strict greedy selection, it produced severe degradation in signal detection across behavioral categories and was excluded from the final configuration. After that, we injected the system-level instructions into the system prompt to ensure strong adherence to the signal definitions.

Let $I$ denote the composite input to the Multimodal LLM. We define $I$ as a tuple sequence:

$$I = (T_{instruct}, T_{transcript}, \{v\}_{i=1}^{N_i})$$

where $T_{instruct}$ represents the static signal definitions, $T_{transcript}$ is the video-specific audio text (transcript), and $\{v\}_{i=1}^{N_i}$ is the sequence of $N$ sampled image frames (where $N$ varies per clip). The model predicts which signals are present and the probability of signal presence $P(y|I)$ based on this unified context.

Behavioral signals were referred from the clinical criteria outlined by (Deng et al., 2024). The static instruction consisted of a generic context, an expected output structure, and a short definition of each of the ten outlined behavioral signals. Each signal was denoted by a shorthand symbol ($S1 - S10$) to reduce the overall token count during model interaction. The output signals identified by the model were later decoded to their original verbatim for readability. The mapping of internal codes to behavioral definitions is as follows:

- **S1:** Absence or Avoidance of Eye Contact - *(Social domain)*
- **S2:** Aggressive Behavior - *(Social domain)*
- **S3:** Hyper- or Hyporeactivity to Sensory Input - *RRB (Restricted and Repetitive Behaviors)*
- **S4:** Non-Responsiveness to Verbal Interaction - *(Social domain)*
- **S5:** Non-Typical Language - *(Social domain)*
- **S6:** Object Lining-Up - *RRB (Restricted and Repetitive Behaviors)*
- **S7:** Self-Hitting or Self-Injurious Behavior - *RRB (Restricted and Repetitive Behaviors)*
- **S8:** Self-Spinning or Spinning Objects - *RRB (Restricted and Repetitive Behaviors)*

- **S9:** Upper Limb Stereotypies - *RRB (Restricted and Repetitive Behaviors)*
- **S10:** Background (i.e., not-applicable)

The behavioral categories for the social domain were identified from the social behavior classifications of the *DSM-V-TR* American Psychiatric Association (2022) and the *M-CHAT-R/F* screening tool (Robins et al., 2009).

**Sample Definition:**

*S1: "Absence or Avoidance of Eye Contact" [A consistent lack of direct gaze or active aversion to maintaining eye contact during social interactions, often limiting nonverbal engagement.]*

*S2: "Aggressive Behavior" [Physical acts of hostility directed outward toward other people or property, often triggered by communication frustration, disrupted routines, or sensory overwhelm.]*

**Output Structure:**

The system produces a structured per-signal report for each video clip, organized as a table with the following fields:

- **Signal:** The behavioral category being evaluated (S1–S10), decoded to its full verbatim label for readability.
- **Observed:** A three-state judgment — Yes (signal detected), No (signal absent), or Unclear (insufficient visual or audio evidence to determine presence). The Unclear state is distinct from No: it reflects epistemic limitation rather than confirmed absence, and is returned when the relevant modality is unassessable from the clip (e.g., no verbal interaction is captured, or camera angle precludes gaze analysis).
- **Confidence:** A categorical tier derived from the frame-voting metric - High (≥70%), Medium (30–69%), or Low ($< 30\%$).
- **Frames Detected:** The explicit frame-voting count expressed as $n/N$ (percentage), where *n* is the number of frames in which the signal was detected and *N* is the total number of frames evaluated for the clip.
- **Note:** A per-signal narrative explaining the visual or contextual basis for the judgment.
- A Clinical Observations section appended below the table provides a holistic narrative summary integrating signal detections, audio transcript content, and behavioral context into a single interpretive statement for reviewer use.

**Frame-Voting Confidence:**

The original model output included a self-reported confidence float. While prior work indicates that large language models can be well-calibrated on text-based factual knowledge Kadavath et al. (2022), our empirical inspection confirmed that the model's raw confidence estimates can be inconsistent across structurally similar clips. To ensure reliability for clinical observation, we replaced the raw float with a computed frame-voting metric. For each video clip with $N$ selected frames, each frame is evaluated independently. If a behavioral signal is detected in $n$ of those $N$ frames ($n \leq N$), the frame-voting confidence is:

$$confidence = n/N$$

This is categorized as: High ($\geq 0.70$), indicating a signal detected across a supermajority of frames and treated as strong evidence. Medium ($0.30$ to $0.69$) denotes a signal present in approximately half the frames, which is flagged for human review. Low ($< 0.30$) means the signal was detected in fewer than one in three frames and is treated as insufficient evidence. We conceptualize the independent frame-level evaluations as analogous to an expert panel seeking consensus.

The $70\%$ threshold is informed by commonly used consensus levels in Delphi method research Diamond et al. (2014); Hsu and Sandford (2007), where agreement thresholds are often predefined and frequently fall in the range of $70\%$ or higher, though no universal standard exists. The $30\%$ lower bound is introduced here as a symmetric heuristic to represent a practical threshold for absence of consensus, while preserving an intermediate range for human adjudication.

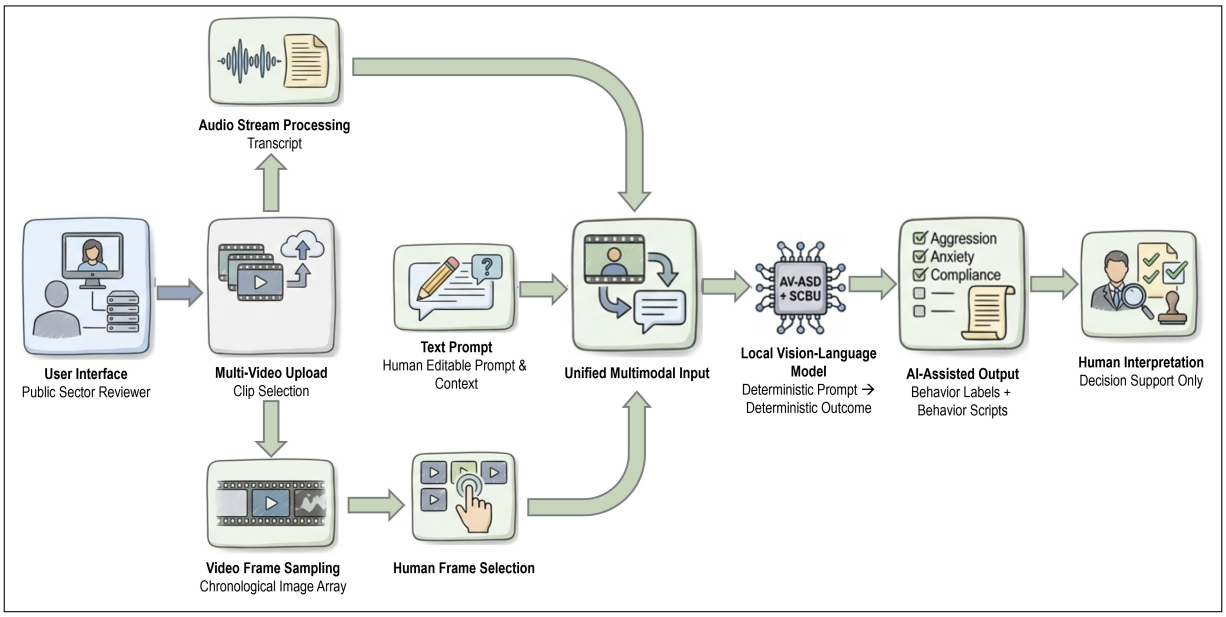



**Fig. 2** – System Workflow

### *3.5. Validation*

To evaluate the performance of the proposed AI-assisted video analysis system, we conducted a systematic comparison between the model-generated outputs and the therapist-annotated ground truth provided in the AV-ASD dataset from the original study (Deng et al., 2024). For each of the video clips, the Multimodal LLM output was normalized into a binary, multi-label representation aligned with the original dataset schema (including the Background class), and each row was identifiable using the corresponding $Video_ID$. A value of 1 indicates the presence of a behavior as identified by the model, while 0 indicates absence. When multiple behaviors were identified within a single clip, all corresponding labels were marked as present. Let $y_{i,k}^{(T)}$ denote the therapist-annotated ground truth label and $y_{i,k}^{(M)}$ denote the model-predicted label for behavioral category $k$ in video clip $i$. Model performance was evaluated by comparing $y^{(M)}$ against $y^{(T)}$ across all of the matching $788$ clips and categories, joined by $Video_ID$. No post hoc filtering or threshold tuning was applied beyond the deterministic model configuration (greedy decoding, fixed seed) described in the Prompt Design subsection. Reported confidence values reflect the frame-voting metric $(n/N)$ rather than the raw LLM-generated float; the Frames Detected column in the output table makes the evidentiary basis for each signal judgment explicit and auditable.

Performance was assessed independently for each behavioral category using standard classification metrics, including accuracy, precision, recall, specificity, F1 score, false positive rate (FPR), and false negative rate (FNR) (Powers, 2020). In addition, behavior prevalence was calculated for each category to provide the necessary context for interpreting these metrics, particularly given the dataset's imbalance. Overall system behavior was computed by pooling predictions across all $10$ behavioral categories (Lordan et al., 2021). This aggregation was intended to characterize the system's general operating profile, specifically, its tendency toward conservative or permissive behavior detection, rather than to suggest a single diagnostic performance measure.

This validation framework reflects the intended role of the system as a decision-support tool rather than an automated classifier. No attempt was made to optimize the model against the evaluation metrics through fine-tuning or prompt adaptation based on validation results. Instead, the evaluation was designed to assess how a fixed, reproducible, and interpretable prompt-driven approach performs when applied to unconstrained, in-the-wild social media video data under conditions that mirror potential public-sector deployment. Figure 2 presents the workflow of the proposed framework.

**Tab. 1** – Performance Metrics

| Behavior | Acc | P | R | Spc | F1 | FPR | FNR | Prev | PP | TP | FP | TN | FN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Absence or Avoidance of Eye Contact | 0.83 | 0.63 | 0.33 | 0.95 | 0.44 | 0.05 | 0.67 | 0.19 | 0.10 | 51 | 30 | 604 | 102 |
| Aggressive Behavior | 0.91 | 0.70 | 0.75 | 0.94 | 0.72 | 0.06 | 0.25 | 0.16 | 0.17 | 92 | 40 | 624 | 31 |
| Hyper- or Hyporeactivity to Sensory Input | 0.83 | 0.51 | 0.21 | 0.96 | 0.30 | 0.04 | 0.79 | 0.18 | 0.07 | 29 | 28 | 621 | 109 |
| Non-Responsiveness to Verbal Interaction | 0.87 | 0.70 | 0.22 | 0.98 | 0.34 | 0.02 | 0.78 | 0.15 | 0.05 | 26 | 11 | 659 | 91 |
| Non-Typical Language | 0.87 | 0.50 | 0.09 | 0.99 | 0.15 | 0.01 | 0.91 | 0.13 | 0.02 | 9 | 9 | 675 | 94 |
| Object Lining-Up | 0.98 | 0.84 | 0.78 | 0.99 | 0.81 | 0.01 | 0.22 | 0.05 | 0.05 | 32 | 6 | 740 | 9 |
| Self-Hitting or Self-Injurious Behavior | 0.88 | 0.65 | 0.16 | 0.99 | 0.26 | 0.01 | 0.84 | 0.13 | 0.03 | 17 | 9 | 674 | 87 |
| Self-Spinning or Spinning Objects | 0.89 | 0.62 | 0.34 | 0.97 | 0.44 | 0.03 | 0.66 | 0.13 | 0.07 | 35 | 21 | 662 | 69 |
| Upper Limb Stereotypies | 0.73 | 0.59 | 0.74 | 0.73 | 0.66 | 0.27 | 0.26 | 0.35 | 0.44 | 203 | 141 | 372 | 71 |
| Background | 0.85 | 0.30 | 0.20 | 0.93 | 0.24 | 0.07 | 0.80 | 0.12 | 0.08 | 19 | 45 | 647 | 76 |
| Overall | 0.86 | 0.60 | 0.41 | - | 0.49 | - | - | - | - | 513 | 340 | 6278 | 739 |

[1] Acc: Accuracy, P: Precision, R: Recall, Spc: Specificity, Prev: Prevalence, PP: Predicted Prevalence

# 4. Findings

### *4.1. Multimodal LLM Classification Performance*

Table 1 summarizes the per-behavior performance of the proposed AI-assisted video analysis system across nine autism-related behavioral categories, one background class, and overall, across all 10 categories. The macro-average F1 score across ten behavioral categories was $0.49$ (precision: $0.60$, recall: $0.41$), reflecting substantial variation in performance across behavior types. Overall accuracy was $0.86$; however, this figure must be interpreted with caution. Because each behavioral class is heavily imbalanced, with negative examples comprising the large majority of observations, a model that predominantly predicts negative would achieve similarly high accuracy without meaningful detection. Per-class F1, precision, recall, and specificity therefore serve as the primary indicators of system utility throughout this analysis.

*Object Lining-Up* achieved the highest F1 score ($0.81$), with high precision ($0.84$) and recall ($0.78$). *Upper Limb stereotypies* also showed relatively strong performance ($F1 = 0.66$). These behaviors tend to be visually salient and structured, which likely makes them easier for the model to capture. From a clinical and service delivery perspective, this indicates reliable automated support for behaviors commonly emphasized in observational screening, potentially reducing manual review burden.

*Absence or Avoidance of Eye Contact* and *Non-Responsiveness to Verbal Interaction* demonstrated moderate performance (F1 = $0.44$ and $0.34$, respectively). Specificity was high ($\geq 0.95$) while recall was lower, indicating conservative detection. In practice, this means fewer false positives in socially sensitive categories, with the system primarily serving to highlight clips that may warrant closer human review.

*Non-Typical Language*, *Hyper- or Hyporeactivity to Sensory Input*, and *Self-Hitting or Self-Injurious Behavior* are more Subtle or context-dependent and exhibited lower recall and F1 scores ($\leq 0.30$), despite high specificity. These results reflect the difficulty of identifying nuanced linguistic, affective, or low-frequency behaviors from short social media clips and underscore the need for continued human interpretation.

The Background class presents a clear illustration of how class imbalance inflates accuracy and specificity: with a prevalence of only $0.12$, the majority of clips are non-background by definition, making high accuracy ($0.85$) and specificity ($0.93$) trivially achievable without meaningful detection. The more informative metrics are F1 ($0.24$) and recall ($0.20$), which reveal that the system correctly identifies only one in five clips that contain no behavioral signal - a limitation that carries practical consequences, as missed Background classifications contribute to the overall false-positive burden across behavioral categories. Across all categories, the system produced substantially more true negatives than false positives, reflecting a consistent bias toward high specificity that must be understood in the context of this structural imbalance rather than as evidence of general classification competence.

### *4.2. Translating System Output to Caregiver-Facing Guidance*

While *Table 1* reports classification metrics calibrated against therapist annotations, a non-specialist user requires a different frame: not how the system performs against a clinical benchmark, but what each output combination means for their next action. The system produces two orthogonal output fields per behavioral category: *Observed (Yes / No / Unclear)* and *Confidence (High / Medium / Low, derived from the frame-voting metric n/N)*. These fields carry distinct interpretive weight and must be read together. *Unclear* is a structurally distinct third state - it does not mean "probably not present." It means the relevant modality was unassessable from the submitted clip (e.g., camera angle precluded gaze analysis; no speech was captured for language categories). Confidence reflects detection consistency across frames and is interpretable only when Observed is *Yes* or *No*; when Observed is *Unclear*, frame-voting confidence does not apply.

Table 2 maps these combinations to recommended caregiver actions.

| Observed | Confidence | System Meaning | Caregiver Guidance |
|---|---|---|---|
| Yes | High ($\geq$70%) | Signal consistently detected across the majority of frames — strong evidentiary basis. | Strong basis to request a professional referral. |
| | Medium (30–69%) | Signal detected in some frames — transient, brief, or partially captured behavior. | Document the clip; monitor for recurrence; consider referral if pattern persists across multiple submissions. |
| | Low ($<$30%) | Signal detected in few frames — weak evidentiary basis. | Note as a data point; do not act on this result alone; continue observation and submit additional clips. |
| No | High ($\geq$70%) | Model found no signal consistently across frames. | **Does not rule out concern.** Overall system recall is 0.41; the model misses more than half of true positive cases. Any persistent caregiver concern should be raised with a healthcare provider regardless of this result. |
| | Low / Medium | Mixed or sparse evidence, net negative judgment. | Treat as equivalent to No / High — absence of a flag is not clinical clearance. |
| Unclear | — | Submitted clip lacked sufficient evidence for this category (e.g., face not visible for eye contact; no speech captured for language categories). | **This is not a negative result.** Submit a new clip that better captures the behavior of concern; see the `Note` field for what the model could not assess. |

**Tab. 2** – Caregiver-facing interpretation of system output. `Observed` and `Confidence` must be read together. `Unclear` is not a negative result and does not carry a confidence tier.

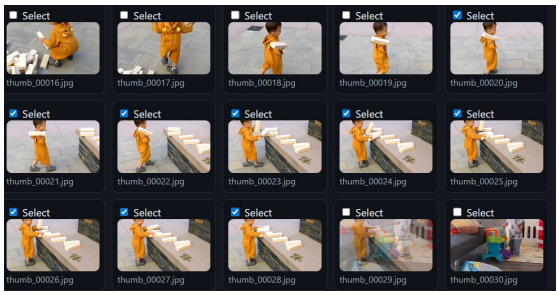
### *4.3. Case-Level Performance Analysis & Model Interpretability*

Below are two representative visual examples for behavioral categories exhibiting contrasting performance profiles, identified by higher and lower F1 scores.

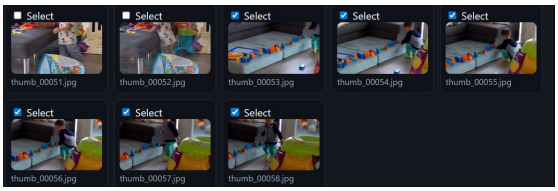
**Object Lining-Up** (highest F1 score) depicts behaviors that are visually salient, spatially structured, and persistent across frames, as seen in both samples. The AI system (model output) and human reviewers (ground truth) arrived at consistent interpretations, supporting reliable flagging of behaviors commonly emphasized in observational screening. For clinicians and public service personnel, this indicates that the system can effectively surface high-confidence cases with minimal ambiguity, reducing manual review effort.

This capacity for clear detection is illustrated in the raw system response for clip *3BQvkkTN3Sk*. As shown below, the model not only flags the signal with high confidence but also provides a precise narrative summary that mirrors clinical observation:

- **Signal:** Object Lining-Up
- **Observed:** Yes
- **Confidence:** High
- **Frames Detected:** 9/9 (100)
- **Note:** Discrete objects are visible and arranged in a lined/stacked pattern in multiple frames.

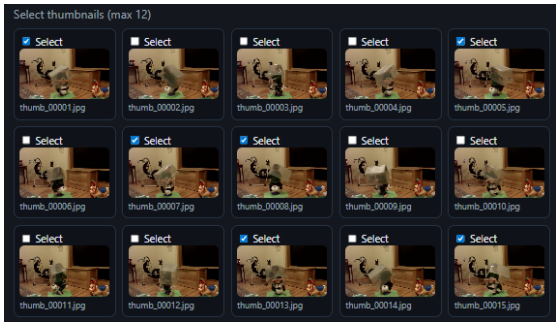


**(a)** Clip 3BQvkkTN3Sk_243_257

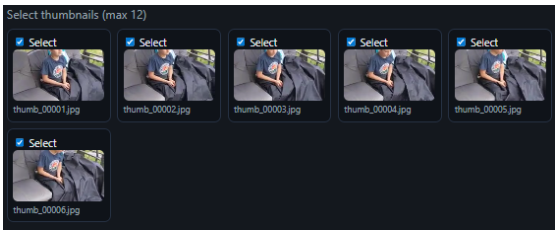


**(b)** Clip 3BQvkkTN3Sk_284_298

**Fig. 3** – Higher F1 score: Object Lining Up



**(a)** Clip -YJhyNoHuUw_0_24



**(b)** Clip 1058358794819229_24_2

**Fig. 4** – Lower F1 score: Hyper-hyporeactivity and Self-hitting

On the contrary, the examples for **Self-Hitting or Self-Injurious Behavior** and **Hyper- or Hyporeactivity to Sensory Input** highlight episodic, context-dependent, or temporally brief behaviors. These categories exhibit the lowest F1 score, lower recall, but high specificity, indicating that the system is cautious in flagging them. In practice, behaviors in these categories tend to be identified only when visual evidence is relatively clear. As a result, false positives are less common in socially and emotionally sensitive cases, though some instances may be missed.

A representative successful detection for this category is shown in the output for clip *1058358794819229_24_27*. The corresponding JSON payload reveals how the model correlates specific visual details - such as the manipulation of fabric - with the internal signal code:

- **Signal:** Self-Hitting or Self-Injurious Behavior
- **Observed:** Yes
- **Confidence:** High
- **Frames Detected:** 6/6 (100%)
- **Note:** The subject exhibits self-hitting behavior across all frames.

When considered alongside the visual examples, the results suggest that the system is better suited to supporting review than to making automated assessments. Behaviors that are visually consistent across frames are more likely to be flagged, while ambiguous cases still depend on human interpretation. This conservative behavior supports scalable review without replacing professional oversight, making the system suitable for use in clinical and public-sector autism screening and service workflows where interpretability and caution are essential.

Addressing the primary research question regarding the system's design for public agency, these performance characteristics, specifically the high specificity ($> 0.90$ for sensitive categories), combined with moderate recall, validate the system's alignment with a "Human-in-Power" framework (Zheng et al., 2024). Unlike "Human-in-the-Loop" systems, where humans merely verify machine outputs, this system places the primary cognitive burden on the user through the manual frame selection mechanism. The model's conservative operating profile ensures it does not override human judgment with frequent false alarms. Instead, it acts as a verification layer that only flags behaviors with strong visual evidence (e.g., Object Lining-Up, $F1 = 0.81$). Consequently, the user retains authority over the "evidentiary process" by curating the visual inputs, while the AI serves strictly to surface high-confidence patterns within that user-defined context. This structural reliance on human input prevents the displacement of professional or caregiver judgment.

## 5. Discussions

In this paper, we aim to examine the effectiveness and governance implications of deploying multimodal LLMs as an early autism screening tool to lower the barrier to professional referral by converting unstructured caregiver held video into structured, auditable signals. Using a prompt-driven, locally deployed framework with human-in-power frame selection, our experiment results indicate that the system reliably classifies visually salient, spatially structured behavioral

categories from user-generated video data while exhibiting conservative, high-specificity behavior for linguistically and contextually complex categories. We further explore the design implications for deploying such a system responsibly in community and public-sector contexts.

Our findings illustrate how the design choices in this framework: local deployment, human-in-power frame selection, and transparent natural-language output, address concrete barriers to deploying citizen-facing AI responsibly in the community and public sector contexts. Specifically, our experimental results demonstrate that user-generated, in-the-wild video can be analyzed by multimodal LLMs with meaningful accuracy for visually salient behavioral categories, suggesting that caregiver-held home video is a viable input modality for structured behavioral review. This finding carries several implications for early autism screening. If caregivers can generate structured behavioral evidence from home video that reliably surfaces high-confidence signals, the initial stage of evidence collection could become decentralized, enabling caregivers to proactively share behavioral observations and potentially assisting clinicians in making earlier referral decisions. Such a workflow could optimize resource allocation by allowing specialists to focus on flagged, high-confidence cases while reducing the informational and logistical barriers that currently delay early intervention. We note that these downstream effects remain hypothetical; our evaluation establishes the technical feasibility and governance architecture of the approach, but confirming its impact on referral timeliness, equity, or caregiver behavior would require longitudinal deployment studies with real-world users.

### *5.1. Public Health and Public Policy Implications*

From a public health and participatory governance perspective, this study demonstrates how multimodal LLMs can be integrated into behavioral review workflows that support early screening referral in ways that expand public participation while maintaining institutional responsibility. By designing the system to let caregivers curate visual evidence and verify model inputs through human-in-the-loop controls, our approach is intended to position citizens as active contributors to early autism behavioral evidence collection rather than passive data sources, offering a pathway toward addressing long-standing gaps between symptom emergence and formal diagnosis. The model's conservative performance profile-reliably identifying visually salient behaviors while maintaining high specificity for socially and emotionally sensitive categories aligns with public health priorities of minimizing harm, reducing unnecessary alarm, and supporting appropriate referral rather than diagnosis. Local deployment and privacy-preserving design further strengthen trust and feasibility in community and public-sector contexts.

However, it is important to note that the classical framework for screening tools prioritizes sensitivity (recall) over specificity, since missed cases (false negatives) in a population-level screener carry greater public health cost than unnecessary referrals (false positives). The current system's high specificity ($> 0.90$ for sensitive categories) is a deliberate governance property: in a public-facing deployment, false alarms erode caregiver trust and generate unnecessary burden on clinical services, so the system is intentionally tuned to flag only when visual evidence is strong. Its limited recall ($0.41$ overall; $0.09$ for *Non-Typical Language*) means it will miss cases—particularly those involving linguistically or contextually complex behaviors—and would be insufficient for a standalone screening instrument. This limitation is deliberate and bounded by the system's design role: it functions as a decision-support layer that augments, rather than replaces, caregiver observation and clinical judgment. For a deployment intended to serve as a primary screener, the operating threshold on the model's confidence output should be tuned to favor recall, accepting a higher false-positive rate in exchange for fewer missed cases. Every other limitation discussed below is bounded by this design role.

### *5.2. Limitations & Future Works*

Within that bounded role, the most consequential implementation constraint is that manual frame selection, while central to the human-in-power design, introduces subjectivity and inter-reviewer variability - reviewer judgment directly shapes model input, which may affect reproducibility across users with varying familiarity with ASD behavioral cues. The system's select-all control partially addresses the scalability concern by enabling full-frame submission when detailed curation is not feasible, though the choice of curation strategy itself remains user-dependent. Beyond this design-level constraint, several technical limitations bound the current evaluation. The study relies on the 928-clip AV-ASD dataset, which, while therapist-annotated, represents a single corpus drawn from social media platforms and limits generalizability to other populations, recording contexts, and cultural settings. The use of 4-bit quantization, while necessary for local deployment on accessible hardware, represents a trade-off between computational feasibility and model precision; its contribution to observed performance gaps, particularly for low-recall categories such as *Non-Typical Language* ($F1 = 0.15$), where subtle linguistic reasoning may be most affected, remains unisolated. The VLM's context window constrains the number of frames that can be submitted per clip, which may affect the detection of brief or low-frequency behaviors in longer recordings. Finally, automatic speech recognition produced incomplete or unusable transcripts in some cases, particularly when audio quality was poor or speech was indistinct, resulting in partial loss of contextual information for language-dependent behavioral categories.

Three directions follow from the limitations and findings of this study. First, future versions could leverage generative AI to automatically identify and extract candidate sub-clips from longer recordings, increasing temporal coverage for brief or low-frequency behaviors while reducing manual review effort. Second, adaptive frame selection strategies dynamically increasing temporal resolution around detected motion or interaction could improve sensitivity to episodic behaviors and represent a natural extension of the human-in-power design. Third, future work should benchmark the quantized model against its full-precision counterpart to isolate the effect of quantization on performance and explore richer audio representations beyond text transcription to better capture affective and communicative cues.

## 6. Conclusions

This study set out to explore whether multimodal LLMs can serve a meaningful role in early autism behavioral review, not as diagnostic instruments, but as tools that help caregivers translate everyday observations into structured evidence worth a clinician's attention. The answer, based on our evaluation, is a qualified yes. The system works well for what it can see clearly and errs on the side of caution for what it cannot, which is precisely the operating profile a public-facing tool in this space should have. What it does not do and should not be expected to do is replace the judgment of caregivers or clinicians. The broader lesson is a design one: when AI is deployed in sensitive public-sector contexts, the question is not only how accurate the model is but also who retains authority over the evidence it processes and how transparent its reasoning is to the people whose lives it touches.

## 7. Contributor Statement

| |
|---|
| Conceptualization: Anirban Pal, Chun-Hua Tsai |
| Methodology: Anirban Pal, Chun-Hua Tsai |
| Software: Anirban Pal |
| Formal Analysis: Anirban Pal |
| Investigation: Anirban Pal |
| Data Curation: Anirban Pal |
| Visualization: Anirban Pal |
| Writing – Original Draft: Anirban Pal |
| Writing – Review & Editing: Ummul Khair Fatima, Srishti Gupta, Pasindu Gayashan, Chun-Hua Tsai |
| Supervision: Chun-Hua Tsai |
| Project Administration: Anirban Pal |

**Tab. 3** – Contributor Statement

## 8. Use of AI

During the preparation of this work, the author(s) used **Claude Code** in order to generate the system code. After using this tool/service, the author(s) reviewed, edited, made the content their own and validated the outcome as needed, and take(s) full responsibility for the content of the publication.

## 9. Conflict Of Interest (COI)

There is no conflict of interest.

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